Matchaya Point of Sale Technical Guide

Matchaya is a CodeIgniter 4 and MySQL point-of-sale application for the IT0049 midterm project. This guide shows how to set up the project, follow the sale and inventory flow, and check the major pages. Its sample café menu and transactions are illustrative.

Set up on Windows with XAMPP

1. Start MySQL in XAMPP and create an empty UTF-8 database named matchaya_pos.
2. Copy .env.example to .env. Check the database host, user, password, port, and the local base URL http://localhost:8084/. Keep .env private.
3. Run composer install from the project directory. Use PHP 8.2 or newer with mysqli, intl, GD, mbstring, fileinfo, and OpenSSL.
4. Run C:\xampp\php\php.exe -d extension=php_intl.dll -d extension=php_gd.dll spark migrate --all.
5. For sample data, run the same PHP command followed by spark db:seed MatchaSeeder. Save the two random staff passwords printed by the seeder.
6. Run Start-Matchaya.cmd and open http://localhost:8084/. Without sample staff, create the first account at /setup. With sample staff, sign in.

Pages

Route	Purpose
/ or /pos	Touch-first menu, cart, and charge
/dashboard	Revenue, orders, top sellers, customer patterns, and stock runway
/products	Product list, form, stock, images, and restock
/customers	Customer records and display-only stamp card
/staff and /account	Staff records and signed-in account
/sales and /sales/new	History, filters, export, and simple no-JavaScript sale
/sales/receipt/{id}	Printable grouped receipt

Database and program flow

Table	Role
products	Menu name, price, current stock, image, and creation time
customers	Optional buyer identity and contact details
users	Staff identity, password hash in password, and optional avatar
sales	One product per line, buyer, staff, quantity, computed total, and time
product_meta	Optional category, description, and featured placement
ci_sessions	Database-backed sessions on Vercel

Routes map each request to a controller. The auth filter checks the small userId session value before any management controller runs. Controllers validate request values, Models and services query or update MySQL, and views display escaped data without database queries.

SaleService begins one transaction. It locks selected products in ID order, rejects an unavailable quantity, applies a conditional stock decrement, computes each total from the saved price, and inserts the sale lines. A failed line rolls the entire order back. Staff identity comes from the session, not the posted form.

The required sales table has no order ID. Multi-item orders share one timestamp, staff ID, and customer ID, and the receipt groups on those values. Two orders by the same staff for the same customer in one second could group together. A real system should add an order identifier when the schema permits it.

Images and access

Product and avatar uploads accept JPEG, PNG, and WebP up to 2 MB and 4096 pixels per side. The server checks actual MIME and dimensions, creates a random name, and prepares 640 by 640 product images or 256 by 256 avatars. Local storage uses GD and writable/uploads; the narrow media route serves generated JPEG names only. The cloud driver uses an incoming Cloudinary transformation when GD is unavailable, so the original is normalized before storage.

The seeded menu has 17 distinct generated product depictions, with local 640 and 1280 pixel WebP versions, descriptive alt text, and a manifest of source, terms, dimensions, and byte sizes. Two brand still lifes support sign-in and About. docs/IMAGE-CREDITS.md and docs/image-contact-sheet.jpg provide the image record and visual sheet. Run tools/check_demo_images.py to verify all committed outputs.

Every management route is protected by the auth filter. Forms include a CSRF field, mutations use POST, passwords are hashed, sessions regenerate after sign-in, and production requires HTTPS. A file-backed login throttle suits local XAMPP; a serverless deployment needs a shared or edge rate limiter.

InfinityFree deployment

The project root .htaccess rewrites requests into public/ within InfinityFree's fixed htdocs web root. Upload the contents of the project, including vendor/, directly into htdocs. Create a new InfinityFree MySQL database and import the seed SQL through its phpMyAdmin. This hosted database is independent of local XAMPP. Put CI_ENVIRONMENT=production, the HTTPS site URL, the exact hosted MySQL hostname and credentials, and MEDIA_DRIVER=local into a private htdocs/.env. See docs/infinityfree.md for the detailed sequence and smoke checks.

The IT0049 sample is live at https://matchaya-it0049.page.gd/. InfinityFree account if0_43139444 uses its own database, if0_43139444_matchaya_pos. A hosted sale, stock decrement, oversell refusal, dashboard, and 640 by 640 uploaded image were checked. docs/deployment-record.md contains the evidence. The local XAMPP stock remained unchanged during the hosted sale.

The optional Vercel path uses the included function entrypoint, hosted MySQL, and MEDIA_DRIVER=cloud with Cloudinary because function storage is ephemeral. Keep .env and database passwords out of GitHub and the submission ZIP.

Troubleshooting

Symptom	Check
Database connection fails	Start MySQL and compare .env host, port, database, user, and password.
Route does not open	Run php spark routes; verify its method, controller, and auth filter.
Image fails	Check image type, GD, or Cloudinary keys.
Image disappears on Vercel	Set MEDIA_DRIVER=cloud and configure Cloudinary credentials.
InfinityFree database fails	Check the assigned host and credentials. Do not use localhost.
Stock is unchanged	Review the sale error. Oversold or invalid carts are rolled back.
No category tabs	Populate product_meta; the core counter still works with All.

Known limitations

This academic schema does not record senior citizen or PWD discounts, VAT exemption, payment method, tips, refunds, voids, or ingredient cost. The dashboard therefore reports revenue and volume, not profit. The customer stamp card is display-only; redemption is not recorded. A real Philippine restaurant deployment must address those legal and operational needs.